

Practice session L8: Particle filtering

Basic information

One purpose with the flipping the classroom model used in this course is to enable us to dedicate our time in class to active learning. We call these sessions our practice sessions. During these sessions we intend to focus on problem solving, peer-instruction, multiple choice questions and discussions on the fundamental concepts that we study. **These classes are for you** – take the opportunity to ask questions and discuss whatever elementary or advanced aspect of this course that you are interested in at the moment.

Today we focus on the material in **lecture 8**. To boost our discussions we have prepared a set of problems that we can look at.

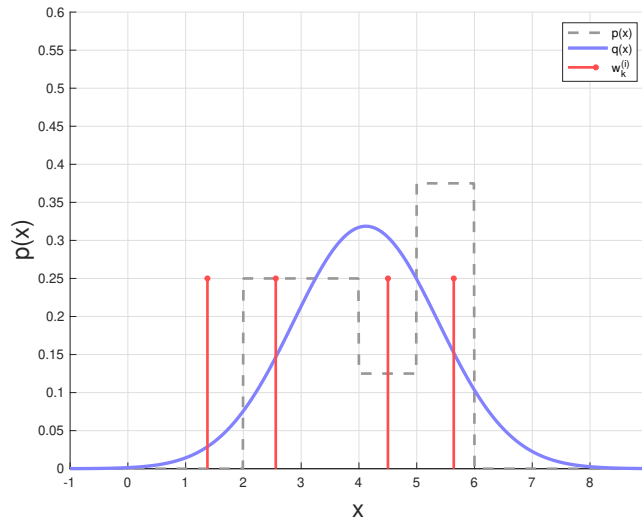


Figure 1: True density $p(x)$, proposal density $q(x)$ and samples from the proposal density.

Problem 1 – Importance sampling

The purpose with this problem is to perform importance sampling in a toy example to get a feeling for what is happening and if it is reasonable.

Suppose that we want to make an Monte Carlo approximation of $p(x)$, as shown in Figure 1, using 4 independent samples. However, we are not able to draw samples from $p(x)$ directly and, hence, we want to use importance sampling to do the approximation. In Figure 1, we can also view our choice of proposal density $q(x)$ and 4 independent samples that we have generated from this proposal density.

- Try to convince each other that $q(x)$ is a reasonable choice for a proposal density. *What are your arguments ?*
- Use the generated samples to make an importance sampling approximation of $p(x)$. Illustrate your approximation together with $p(x)$ and $q(x)$ in Figure 2. *How are the weights of the samples changed for the samples to represent $p(x)$ instead of $q(x)$? Try to convince each other, why this is reasonable.*

Note: You do not have to calculate the weights exactly but make an effort to express them numerically. You will need them in the next problem.

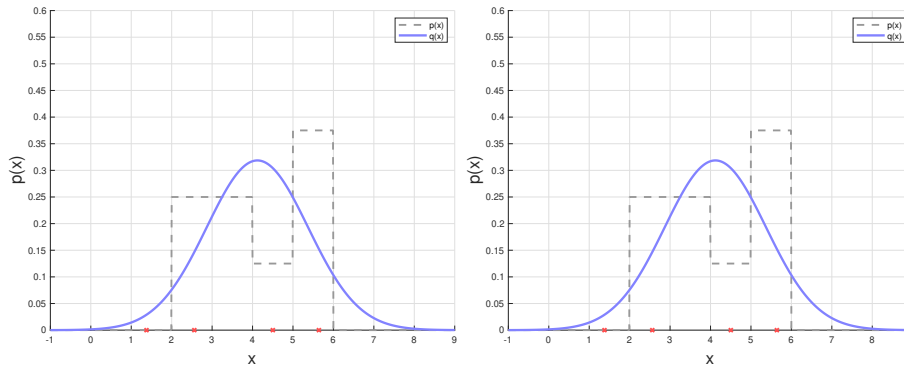


Figure 2: Feel free to illustrate your results here. True density $p(x)$, proposal density $q(x)$.

Problem 2 – Resampling

The purpose with this problem is to perform resampling using the same toy example as above to get a feeling for what is happening and if it is reasonable.

We now want to perform resampling in order to use our samples more efficiently. To do so we have drawn the following numbers from a $\mathcal{U}[0, 1]$ distribution, .54, .41, .67 and .07.

- Agree in the group how you can use the above numbers to perform resampling.
- Use the numbers to perform resampling on the samples that you generated in the previous problem and illustrate the new samples and $p(x)$ in Figure 2. *Agree within the group, in what way is this a better or worse approximation of $p(x)$?*

Note: In order to make the result comparable it could be a good idea to order the samples in ascending order.